HW12

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Exercise 12.1

设 M 是一个光滑的 n 维流形, A 是 M 上的一个光滑的协变 k 阶张量场. 如果 $(U,(x^i))$ 和 $(\tilde{U},(\tilde{x}^j))$ 是 M 上两个重叠的光滑坐标卡, 即 A 可分别写成

$$A = A_{i_1\dots i_k} dx^{i_1} \otimes \cdots \otimes dx^{i_k} = \tilde{A}_{j_1\dots j_k} d\tilde{x}^{j_1} \otimes \cdots \otimes d\tilde{x}^{j_k}.$$

试用 $\tilde{A}_{j_1\dots j_k}$ 表示分量函数 $A_{i_1\dots i_k}$.

Proof. 由坐标切向量的替换公式

$$\frac{\partial}{\partial x^i} = \sum_{p=1}^{n} \frac{\partial \tilde{x}^p}{\partial x^i} \frac{\partial}{\partial \tilde{x}^p}, \quad i = 1, \cdots, n$$

将 A 作用在 $\frac{\partial}{\partial x^{i_1}}, \cdots, \frac{\partial}{\partial x^{i_k}}$ 上, 得到

$$\begin{aligned}
A_{i_1\dots i_k} &= A\left(\frac{\partial}{\partial x^{i_1}}, \cdots, \frac{\partial}{\partial x^{i_k}}\right) \\
&= \sum_{j_1,\cdots,j_k} \tilde{A}_{j_1\dots j_k} d\tilde{x}^{j_1} \otimes \cdots \otimes d\tilde{x}^{j_k} \left(\sum_{p_1=1}^{n} \frac{\partial \tilde{x}^{p_1}}{\partial x^{i_1}} \frac{\partial}{\partial \tilde{x}^{p_1}}, \cdots, \sum_{p_k=1}^{n} \frac{\partial \tilde{x}^{p_k}}{\partial x^{i_k}} \frac{\partial}{\partial \tilde{x}^{p_k}}\right) \\
&= \sum_{j_1,\cdots,j_k} \frac{\partial \tilde{x}^{j_1}}{\partial x^{i_1}} \cdots \frac{\partial \tilde{x}^{j_k}}{\partial x^{i_k}} \tilde{A}_{j_1\dots j_k}
\end{aligned}$$

$\square$

Exercise 12.2 ▶ 偶数维欧氏空间上的典则辛形式

设 $(x^1, \ldots, x^n, y^1, \ldots, y^n)$ 为 $\mathbb{R}^{2n}$ 中的坐标函数, 令

$$\omega = \sum_{i=1}^{n} dx^i \wedge dy^i.$$

1. 计算 $d\omega$.
2. 计算 $\omega^n = \omega \wedge \cdots \wedge \omega$.
3. 计算 $\iota^*\omega$:
   (a) 设 $\iota : \mathbb{R}^{2n-2} \hookrightarrow \mathbb{R}^{2n}$ 为由 $x^n = y^n = 0$ 定义的嵌入.
   (b) 设 $\iota : \mathbb{R}^{n} \hookrightarrow \mathbb{R}^{2n}$ 为由 $y^1 = \cdots = y^n = 0$ 定义的嵌入.
   (c) 设 $\iota : \mathbb{T}^{n} \hookrightarrow \mathbb{R}^{2n}$ 为由 $(x^i)^2 + (y^i)^2 = 1, 1 \leqslant i \leqslant n$ 定义的嵌入.
4. 给定 $f \in C^\infty(\mathbb{R}^{2n})$, 构造 X_f 使得 $\iota_{X_f}\omega = df$.
5. 令 X_f 如上定义. 求证: $\mathcal{L}_{X_f} f = 0$ 且 $\mathcal{L}_{X_f}\omega = 0$.

Proof. 1.

$$d\left(dx^i \wedge dy^i\right) = \left(d\left(dx^i\right)\right) \wedge dy^i - dx^i \wedge d\left(dy^i\right) = 0$$

$$d\omega = \sum_{i=1}^{n} d\left(dx^i \wedge dy^i\right) = 0$$

2. 由于 $dx^i \wedge dy^i$ 是偶形式, 它们之间整体换序不改变符号

$$\begin{aligned}
\omega^n &= \sum_{i_1,\cdots,i_n} dx^{i_1} \wedge dy^{i_1} \wedge \cdots \wedge dx^{i_n} \wedge dy^{i_n} \\
&= \sum_{(i_1,\cdots,i_n)\in S_n} dx^{i_1} \wedge dy^{i_1} \wedge \cdots dx^{i_n} \wedge dy^{i_n} \\
&= n! dx^1 \wedge dy^1 \wedge \cdots \wedge dx^n \wedge dy^n \\
&= (-1)^{\frac{n(n-1)}{2}} n! dx^1 \wedge \cdots \wedge dx^n \wedge dy^1 \wedge \cdots \wedge dy^n
\end{aligned}$$

3. (a)

$$\begin{aligned}
\iota^*\omega &= \sum_{i=1}^{n} d\left(x^i \circ \iota\right) \wedge d\left(y^i \circ \iota\right) \\
&= \sum_{i=1}^{n-1} dx^i \wedge dy^i + d(0) \wedge d(0) \\
&= \sum_{i=1}^{n-1} dx^i \wedge dy^i
\end{aligned}$$

(b)

$$\begin{aligned}
\iota^*\omega &= \sum_{i=1}^{n} d\left(x^i \circ \iota\right) \wedge d\left(y^i \circ \iota\right) \\
&= \sum_{i=1}^{n} d\left(x^i\right) \wedge d(0) \\
&= 0
\end{aligned}$$

(c) 设 $\mathbb{T}^n$ 的坐标为

$$\left(\theta^1, \cdots, \theta^n\right)$$

则 ι 的坐标表示为

$$\left(\theta^1, \cdots, \theta^n\right) \mapsto \left(\cos\theta^1, \sin\theta^1, \cdots, \cos\theta^n, \sin\theta^n\right)$$

此时

$$\begin{aligned}
\iota^*\omega &= \sum_{i=1}^{n} d\left(x^i \circ \iota\right) \wedge d\left(y^i \circ \iota\right) \\
&= \sum_{i=1}^{n} \left(\sum_{j_1} \frac{\partial \cos\theta^i}{\partial \theta^{j_1}} d\theta^{j_1}\right) \wedge \left(\sum_{j_2} \frac{\partial \sin\theta^i}{\partial \theta^{j_2}} d\theta^{j_2}\right) \\
&= \sum_{i=1}^{n} \left(-\sin\theta^i d\theta^i\right) \wedge \left(\cos\theta^i d\theta^i\right) = \sum_{i=1}^{n} \left(-\sin\theta^i \cos\theta^i\right) d\theta^i \wedge d\theta^i \\
&= 0
\end{aligned}$$

4.

$$df = \frac{\partial f}{\partial x^i} dx^i + \frac{\partial f}{\partial y^i} dy^i$$

设

$$X_f = X^i \frac{\partial}{\partial x^i} + Y^i \frac{\partial}{\partial y^i}$$

则

$$\begin{aligned}
\frac{\partial f}{\partial x^j} = \iota_{X_f}\omega\left(\frac{\partial}{\partial x^j}\right) &= \omega\left(X^i \frac{\partial}{\partial x^i} + Y^i \frac{\partial}{\partial y^i}, \frac{\partial}{\partial x^j}\right) \\
&= \sum_{k=1}^{n} dx^k \wedge dy^k \left(X^i \frac{\partial}{\partial x^i} + Y^i \frac{\partial}{\partial y^i}, \frac{\partial}{\partial x^j}\right) \\
&= -Y^j
\end{aligned}$$

因此 $Y^j = -\frac{\partial f}{\partial x^j}$

$$\begin{aligned}
\frac{\partial f}{\partial y^j} = \iota_{X_f}\omega\left(\frac{\partial}{\partial y^j}\right) &= \omega\left(X^i \frac{\partial}{\partial x^i} + Y^j \frac{\partial}{\partial y^i}, \frac{\partial}{\partial y^j}\right) \\
&= \sum_{k=1}^{n} dx^k \wedge dy^k \left(X^i \frac{\partial}{\partial x^i} + Y^i \frac{\partial}{\partial y^i}, \frac{\partial}{\partial y^j}\right) \\
&= X^j
\end{aligned}$$

因此 $X^j = \frac{\partial f}{\partial y^j}$, 故

$$X_f = \sum_{i=1}^{n} \frac{\partial f}{\partial y^i} \frac{\partial}{\partial x^i} - \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} \frac{\partial}{\partial y^i}$$

5. 由 Cartan's Magic Formula 和 4.

$$\begin{aligned}
\mathcal{L}_{X_f} f &= \iota_{X_f}(df) + d\left(\iota_{X_f} f\right) \\
&= \iota_{X_f}(df) \\
&= \iota_{X_f}\left(\iota_{X_f}\omega\right) = \omega\left(X_f, X_f\right) \\
&= 0
\end{aligned}$$

由 Cartan's Magic Formula 和 1.4.

$$
\begin{aligned}
\mathcal{L}_{X_f}\omega &= \iota_{X_f}(d\omega) + d\left(\iota_{X_f}\omega\right) \\
&= \iota_{X_f}(0) + d(df) \\
&= 0
\end{aligned}
$$

□

Exercise 12.3

在 $\mathbb{R}^3$ 上定义一个如下的 2-形式 ω,

$$\omega = xdy \wedge dz + ydz \wedge dx + zdx \wedge dy.$$

a). 在球面坐标 (ρ, φ, θ) 下计算 ω, 其中

$$(x, y, z) = (\rho \sin\varphi \cos\theta, \rho \sin\varphi \sin\theta, \rho \cos\varphi).$$

b). 分别在直角坐标和球坐标下计算 $d\omega$, 并验证这两个表达式表示同一个 3-形式.

c). 设 $\iota_{\mathbb{S}^2} : \mathbb{S}^2 \hookrightarrow \mathbb{R}^3$ 为自然嵌入, 在坐标 (φ, θ) 下计算拉回 $\iota_{\mathbb{S}^2}^*\omega$ (在坐标有定义的开集上).

d). 求证 $\iota_{\mathbb{S}^2}^*\omega$ 在 $\mathbb{S}^2$ 上处处非零.

Proof. 1. 将 (x, y, z) 和 (ρ, φ, θ) 视为 $\mathbb{R}^3$ 上的两组坐标, 则令 $f : \mathbb{R}^3 \to \mathbb{R}^3$,

$$(x, y, z) \mapsto (\rho \sin\varphi \cos\theta, \rho \sin\varphi \sin\theta, \rho \cos\varphi)$$

则 f 实际上是恒等映射, 上述是恒等变换的一个坐标表示. 那么

$$
\begin{aligned}
\omega &= f^*\omega \\
&= (x \circ f)\, d(y \circ f) \wedge d(z \circ f) + (y \circ f)\, d(z \circ f) \wedge d(x \circ f) + (z \circ f)\, d(x \circ f) \wedge d(\\
&= \rho \sin\varphi \cos\theta d(\rho \sin\varphi \sin\theta) \wedge d(\rho \cos\varphi) + \rho \sin\varphi \sin\theta d(\rho \cos\varphi) \wedge d(\rho \sin\varphi \text{co} \\
&\quad + \rho \cos\varphi d(\rho \sin\varphi \cos\theta) \wedge d(\rho \sin\varphi \sin\theta)
\end{aligned}
$$

其中

$$d(\rho \sin\varphi \sin\theta) = \sin\varphi \sin\theta d\rho + \rho \sin\theta \cos\varphi d\varphi + \rho \sin\varphi \cos\theta d\theta$$

$$d(\rho \sin\varphi \cos\theta) = \sin\varphi \cos\theta d\rho + \rho \cos\varphi \cos\theta d\varphi - \rho \sin\varphi \sin\theta d\theta$$

$$d(\rho \cos\varphi) = \cos\varphi d\rho - \rho \sin\varphi d\varphi$$

$$d(\rho\sin\varphi\sin\theta)\wedge d(\rho\cos\varphi)$$
$$=-\rho\sin\theta d\rho\wedge d\varphi+\rho^2\sin^2\varphi\cos\theta d\varphi\wedge d\theta+\rho\sin\varphi\cos\varphi\cos\theta d\theta\wedge d\rho$$
$$d(\rho\cos\varphi)\wedge d(\rho\sin\varphi\cos\theta)$$
$$=\rho\cos\theta d\rho\wedge d\varphi+\rho^2\sin^2\varphi\sin\theta d\varphi\wedge d\theta+\rho\sin\varphi\cos\varphi\sin\theta d\theta\wedge d\rho$$
$$d(\rho\sin\varphi\cos\theta)\wedge d(\rho\sin\varphi\sin\theta)$$
$$=\rho^2\sin\varphi\cos\varphi d\varphi\wedge d\theta-\rho\sin^2\varphi d\theta\wedge d\rho$$

全部带入，得到

$$\begin{aligned}\omega&=\left(-\rho^2\sin\varphi\sin\theta\cos\theta+\rho^2\sin\varphi\sin\theta\cos\theta\right)d\rho\wedge d\varphi\\&+\left(\rho^3\sin^3\varphi\cos^2\theta+\rho^3\sin^3\varphi\sin^2\theta+\rho^3\sin\varphi\cos^2\varphi\right)\mathrm{d}\varphi\wedge\mathrm{d}\theta\\&+\left(\rho^2\sin^2\varphi\cos\varphi\cos^2\theta+\rho^2\sin^2\varphi\cos\varphi\sin^2\theta-\rho^2\sin^2\varphi\cos\varphi\right)\mathrm{d}\theta\wedge\mathrm{d}\rho\\&=\rho^3\sin\varphi d\varphi\wedge d\theta\end{aligned}$$

因此

$$\omega=\rho^3\sin\varphi d\varphi\wedge d\theta$$

2.

$$\begin{aligned}d\omega&=d(xdy\wedge dz)+d(ydz\wedge dx)+z(dx\wedge dy)\\&=dx\wedge dy\wedge dz+dy\wedge dz\wedge dx+dz\wedge dx\wedge dy\\&=3dx\wedge dy\wedge dz\end{aligned}$$

$$\begin{aligned}d\omega&=d\left(\rho^3\sin\varphi\right)\wedge d\varphi\wedge d\theta\\&=\left(3\rho^2\sin\varphi d\rho+\rho^3\cos\varphi d\varphi\right)\wedge d\varphi\wedge d\theta\\&=3\rho^2\sin\varphi d\rho\wedge d\varphi\wedge d\theta\end{aligned}$$

$$\begin{aligned}&3dx\wedge dy\wedge dz\\&=f^*(3dx\wedge dy\wedge dz)\\&=3d(\rho\sin\varphi\cos\theta)\wedge d(\rho\sin\varphi\sin\theta)\wedge d(\rho\cos\varphi)\\&=3(\rho^2\sin^3\varphi\sin^2\theta+\rho^2\sin^3\varphi\cos^2\theta\\&+\rho^2\sin\varphi\cos^2\varphi\cos^2\theta+\rho^2\sin\varphi\cos^2\varphi\sin^2\theta)d\rho\wedge d\varphi\wedge d\theta\\&=3\rho^2\sin\varphi d\rho\wedge d\varphi\wedge d\theta\end{aligned}$$

3. $\iota_{\mathbb{S}^2}$ 的坐标表示为

$$\iota_{\mathbb{S}^2}(\varphi,\theta)=(1,\varphi,\theta)$$

$$
\begin{aligned}
\iota_{\mathbb{S}^2}^*\omega &= (\rho\circ\iota)^3 \sin(\varphi\circ\iota)\, d(\varphi\circ\iota)\wedge d(\theta\circ\iota) \\
&= \sin\varphi d\varphi\wedge d\theta
\end{aligned}
$$

4. 在坐标开集 $\{\varphi\in(0,\pi)\}$ 上, 恒有 $\sin\varphi\neq 0$, 因此 $\iota_{\mathbb{S}^2}^*\omega$ 在其上非退化. 注意到 ω 是关于 (x,y,z) 轮换对称的, 我们建立新坐标

$$(x',y',z')=(y,z,x)$$

则

$$\omega = x'dy'\wedge dz' + y'dz'\wedge dx' + z'dx'\wedge dy'$$

对 (x',y',z') 建立相应的球面坐标 (ρ',φ',θ')

$$(x',y',z')=(\rho'\sin\varphi'\cos\theta',\rho'\sin\varphi'\sin\theta',\rho'\cos\varphi')$$

可以类似地得到

$$\iota_{\mathbb{S}^2}^*\omega = \sin\varphi' d\varphi'\wedge d\theta'$$

则在坐标开集 $\{\varphi'\in(0,\pi)\}$ 上, $\iota_{\mathbb{S}^2}^*\omega$ 非退化. 注意到

$$\{\varphi\in(0,\pi)\}\cup\{\varphi'\in(0,\pi)\}=\mathbb{S}^2$$

因此 $\iota_{\mathbb{S}^2}^*\omega$ 在 $\mathbb{S}^2$ 上非退化.

$\square$